STUDENT PERFORMANCE DATA ANALYSIS

Exploratory Analysis and Visualisation of Examination Scores

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Course	B.Sc CS (AI & DS)
Dataset Used	Students Performance Dataset
Tools Used	Python, Pandas, NumPy, Matplotlib and Seaborn

A data-driven study of student scores, preparation patterns and relationships among subjects.

1. Objective

The objective of this report is to explore student examination performance in Mathematics, Reading and Writing. The analysis examines overall score distributions, pass/fail outcomes, parental education categories, test-preparation status and correlations among subject scores. The findings are intended to describe patterns in the supplied dataset and support evidence-informed academic planning.

2. Dataset Overview

The notebook loads a Students Performance CSV dataset containing 1,000 student records and 8 original variables. The fields cover student gender, race/ethnicity group, parental education, lunch category, test-preparation course status and scores in three subjects.

Feature	Value	Interpretation
Student records	1,000	Rows available in the dataset
Original variables	8	Five categorical and three numerical fields
Score subjects	3	Math, Reading and Writing
Null values	0	No missing values reported in the notebook
Duplicate rows	0	No duplicate records reported
Test preparation	2 categories	Completed and None

The dataset contains complete values for all eight columns according to the notebook's **info()** and **isnull().sum()** outputs. The duplicate check also returned zero.

3. Data Inspection and Preprocessing

The notebook uses Pandas to load and inspect the CSV file. The first and last records are viewed with **head()** and **tail()**, while **shape**, **columns**, **info()** and **describe()** are used to understand the dataset structure and score ranges.

The following preparation steps are present in the notebook:

• Column names are stripped of extra spaces and converted to lowercase.
• Longer column names are renamed to concise labels such as **parental_education**, **test_prep** and **math_score**.
• Text values in categorical columns are stripped and standardized using title case.
• The three subject score columns are selected for numerical summaries and visual analysis.

Derived performance measures

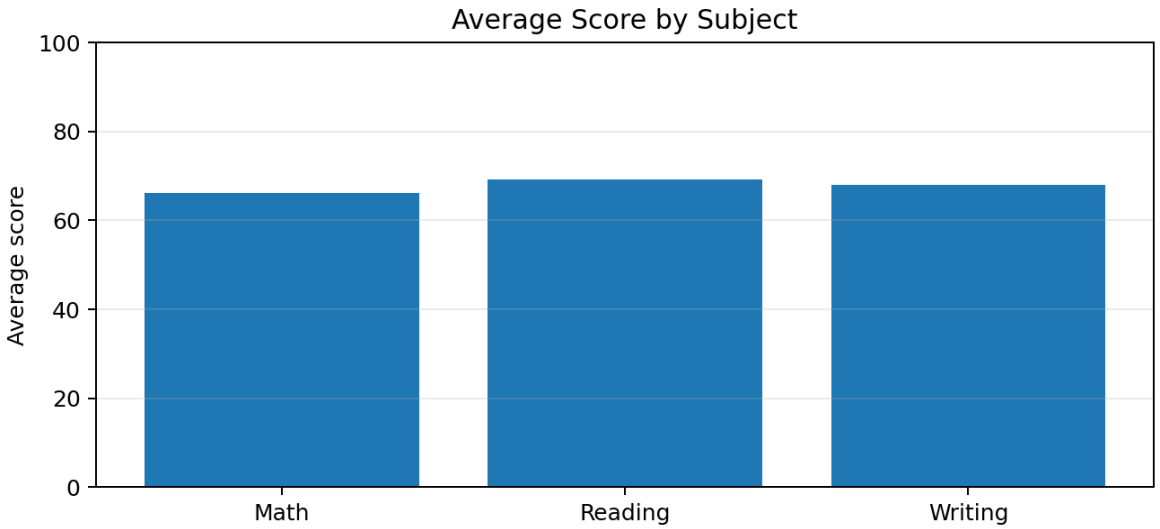
The notebook calculates each student's total score by adding the three subject marks. It then calculates average score and percentage using the maximum combined score of 300. A student is classified as Pass only when all three subject scores are at least 40; otherwise the result is Fail.

Derived field	Method
Total score	Math + Reading + Writing
Average score	Total score ÷ 3
Percentage	(Total score ÷ 300) × 100
Result	Pass if all three subject scores ≥ 40; otherwise Fail

> This report follows the pass/fail rule defined in the uploaded notebook; it does not substitute a different grading policy.

4. Overall Subject Performance

Descriptive statistics show that Reading has the highest average score, followed by Writing and Mathematics. The median scores show a similar pattern, with Reading at 70, Writing at 69 and Mathematics at 66.

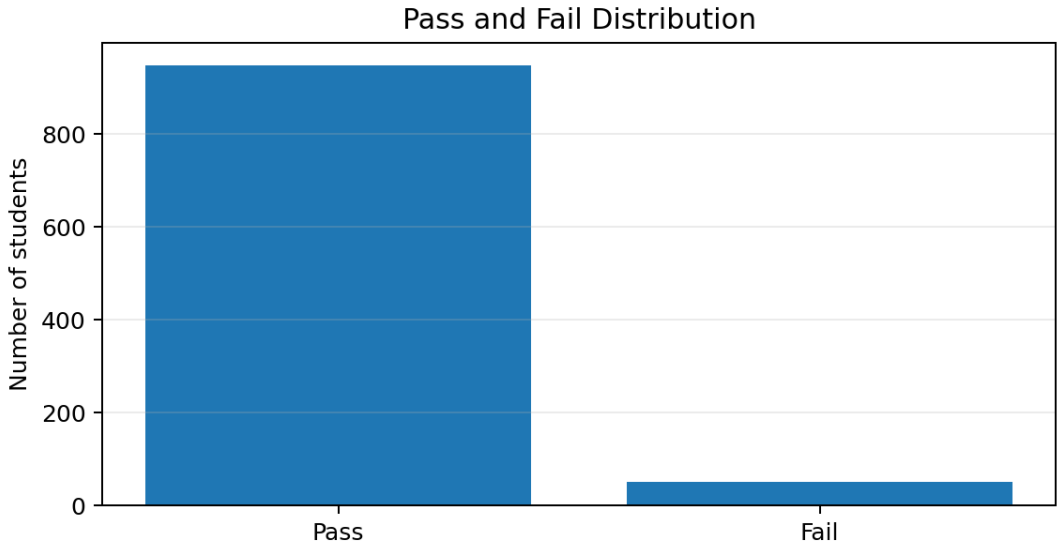


Subject	Mean	Median	Standard deviation
Math	66.09	66	15.16
Reading	69.17	70	14.60
Writing	68.05	69	15.20

Observation: Reading has the strongest average performance in this dataset (69.17). Mathematics has the lowest mean (66.09). The standard deviations are close to 15 marks, indicating a noticeable spread of student scores in each subject.

5. Pass/Fail and Outlier Analysis

The notebook defines a passing result as achieving at least 40 marks in each of the three subjects. Under this rule, 949 students are classified as Pass and 51 as Fail.



Result	Students	Percentage
Pass	949	94.9%
Fail	51	5.1%
Total	1,000	100%

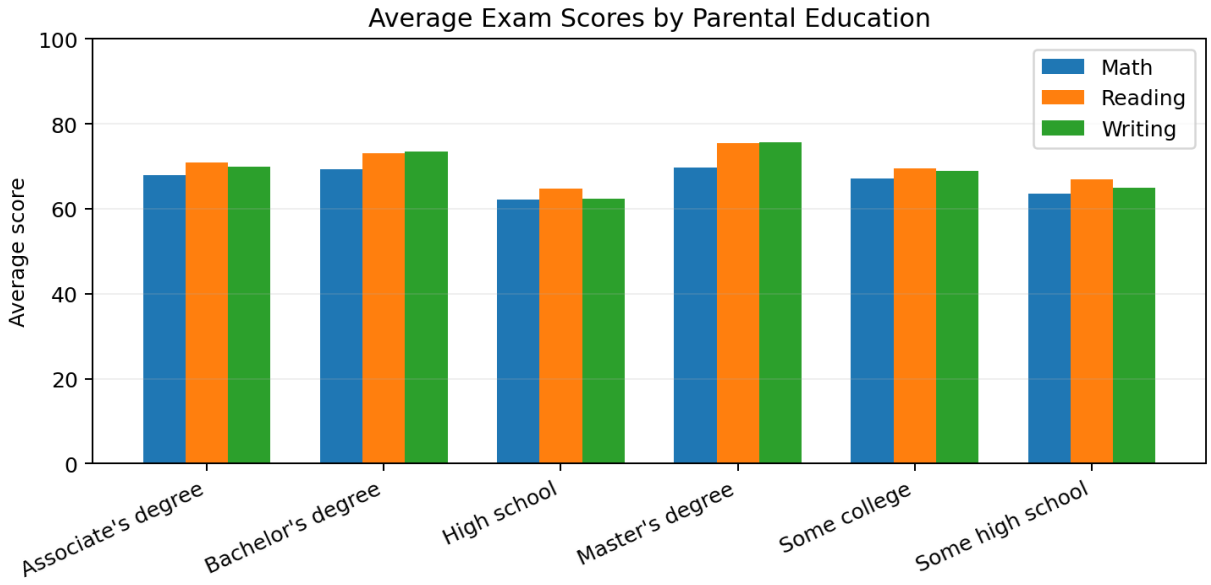
Potential outliers were identified using the Interquartile Range (IQR) rule. The notebook reports 8 potential outliers in Mathematics, 6 in Reading and 5 in Writing. These are flags for review and are not automatically errors.

Subject	Potential outliers
Math	8
Reading	6
Writing	5

Observation: The pass proportion is 94.9% under the notebook's three-subject threshold. A small number of unusually low or high observations may merit review, while preserving valid student performance values.

6. Performance by Parental Education

The notebook groups students by parental education level and calculates the average marks in Mathematics, Reading and Writing for each group. The grouped bar chart compares these subject averages across six education categories.

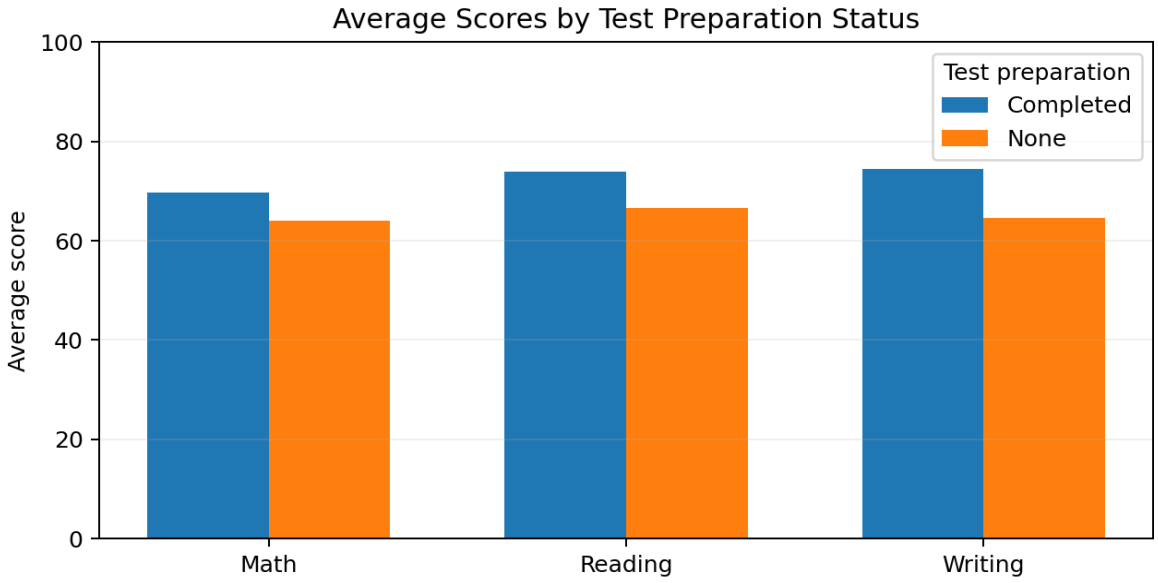


Parental education	Math	Reading	Writing
Associate's degree	67.88	70.93	69.90
Bachelor's degree	69.39	73.00	73.38
High school	62.14	64.70	62.45
Master's degree	69.75	75.37	75.68
Some college	67.13	69.46	68.84
Some high school	63.50	66.94	64.89

Observation: Students in the Master's degree parental-education category have the highest average scores in Reading (75.37) and Writing (75.68), as well as a Math average of 69.75. The High school category has the lowest average in all three subjects in this summary. These are descriptive group differences and do not establish that parental education causes score differences.

7. Test Preparation Analysis

The notebook compares subject scores for students who completed the test-preparation course and those recorded as having no completed course. The dataset contains 358 students in the Completed category and 642 in the None category.

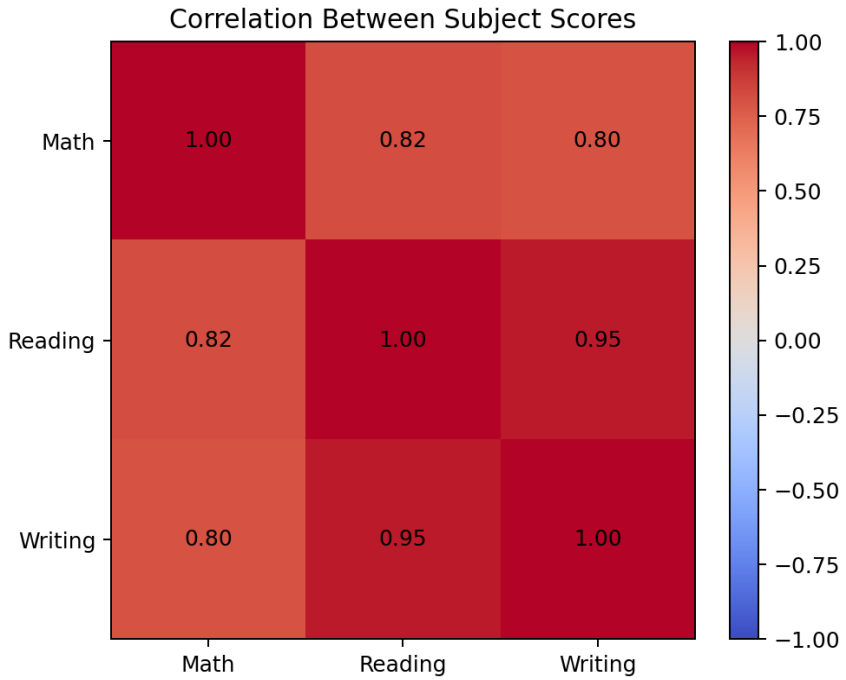


Subject	Completed	None	Difference
Math	69.70	64.08	5.62
Reading	73.89	66.53	7.36
Writing	74.42	64.50	9.91

Observation: The completed-preparation group has higher average scores in all three subjects. The largest difference is in Writing (about 9.91 marks), followed by Reading (7.36) and Mathematics (5.62). This comparison shows an association in the dataset; it does not prove that the course alone caused the higher scores.

8. Subject Relationship and Correlation Analysis

Pearson correlation is used in the notebook to measure the strength and direction of linear relationships among Mathematics, Reading and Writing scores. The resulting correlation matrix is visualised using a heatmap.



Subject pair	Pearson correlation	Interpretation
Math – Reading	0.8176	Strong positive relationship
Math – Writing	0.8026	Strong positive relationship
Reading – Writing	0.9546	Very strong positive relationship

Observation: All three subject pairs show positive correlations. Reading and Writing have the strongest relationship (0.9546). Students with higher scores in one subject tend to also score higher in the other, especially for Reading and Writing. Correlation does not establish causation.

9. Key Findings and Recommendations

Key Findings

• The dataset contains 1,000 records with no missing values and no duplicate rows reported.
• Reading has the highest subject mean (69.17), while Mathematics has the lowest (66.09).
• 949 students (94.9%) pass under the notebook's rule requiring at least 40 in all three subjects.
• The Master's degree parental-education group has the highest average Reading and Writing scores in the grouped summary.
• Students who completed test preparation have higher average marks in all three subjects.
• Reading and Writing scores have a very strong positive correlation of 0.9546.

Recommendations

1. Subject-focused support: Review Mathematics performance and provide additional practice or academic support where needed, while continuing to strengthen Reading and Writing skills.

2. Targeted student support: Use the pass/fail results to identify students who may benefit from additional learning support. Review individual subject marks rather than relying only on the overall average.

3. Encourage preparation: The completed-preparation group shows higher averages across all subjects. Schools may evaluate access to structured preparation opportunities for all students.

4. Use data carefully: Treat group comparisons and correlations as descriptive evidence. Consider other relevant factors before making decisions about student outcomes.

The notebook also creates a comparison chart for lunch category, but the saved output does not include the underlying group-average table. Therefore, no numerical lunch-group conclusion is stated in this report.

10. Conclusion

This analysis presents an overview of student performance using descriptive statistics, derived score measures, group comparisons and visualisations. The dataset includes 1,000 student records and reports no missing values or duplicate rows. Reading records the highest average mark, while Mathematics has the lowest average among the three subjects.

Using the pass rule defined in the notebook, 94.9% of students are classified as Pass. The grouped analysis indicates higher average scores among students who completed the test-preparation course. The parental-education comparison also shows differences in average scores across categories. Correlation analysis identifies a particularly strong positive relationship between Reading and Writing scores.

Overall, the findings can help describe learning patterns and identify areas for further academic support. However, the results are observational and should not be interpreted as proof of cause and effect. Further analysis using additional student-level and contextual information may provide a more complete understanding of performance.

Tools and Libraries Used

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

— End of Report —